JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 95, NO. B12, PAGES 19,185–19,198, NOVEMBER 10, 1990

A Devonian Paleomagnetic Pole From Red Beds of the Tarim Block, China

YIANPING LI[1], MICHAEL MCWILLIAMS[1], ROBERT SHARPS[1], ALLAN COX[1,2]
YONGAN LI[3], QIANG LI[3], ZHENGJIA GAO[3], ZHENGKUN ZHANG[4], YONGJIAN ZHAI[5]

We present new Devonian paleomagnetic results from 59 sites in three stratigraphic sections exposed on the northwestern margin of the Tarim Block. At one section, 1998 m of red sandstone is continuously exposed; the other two sections can be correlated with the first on both magnetostratigraphic and lithologic grounds. Progressive thermal demagnetization reveals three characteristic magnetizations. One is a Late Permian overprint which is isolated below 578°C at sites near contacts with Late Permian dikes. The other two components are approximately antipodal, have a much higher unblocking temperature than the overprint, and are directionally distinct from the Late Permian overprint. We believe that these are characteristic Devonian magnetizations. Within the continuous 1998 m section, our results denote a 395 m reversed polarity zone overlying a 1603 m normal polarity zone. One reversed event is recorded at the base of the section. A paleomagnetic pole calculated by averaging results from 47 normal and reversed sites lies at $\lambda_p = 16.5°N$, $\phi_p = 165.0°E$, $K = 25$, and $A_{95} = 4.3°$. This pole is statistically distinct from a previously reported Late Devonian pole for Tarim. On the basis of field geologic and rock magnetic studies, we believe that the previously reported pole, in addition to results from one of our three sections, reflects at least partial contamination in the form of a thermal overprint caused by Late Permian igneous activity. An analysis of Devonian to Late Carboniferous polar wander suggests that the Tarim Block was attached to a subducting plate, and moved northward and rotated clockwise during the late Paleozoic. Comparison of the Devonian paleolatitudes of Siberian, Kazakhstan, Tarim and South China Blocks indicates that both the Tarim and South China Blocks were located in equatorial regions and were far south of the northern Angaran plate (Siberia and Kazakhstan).

Introduction

Western China consists primarily of the Tarim and Junggar blocks to the north and numerous accreted terranes to the south. The Tarim Block lies between the Tian Shan Range to the north and west, and the Kunlun-Altyn Mountains to the south. Field evidence suggests that the Tarim Block collided with the Tian Shan arc in Carboniferous to Permian times [*Wang and Liu*, 1981; *Zhang et al.*, 1984; *Chen and Liang*, 1985; *Zhang and Wu*, 1985; *Watson et al.*, 1987; *Hu*, 1985; *Smith*, 1988; *Gao and Wu*, 1984; *Ji and Coney*, 1985]. Because the Tarim Block is the most inboard block of China, an understanding of its motion is important in constructing the history of terrane accretion to the south.

Permian, Carboniferous and Devonian paleomagnetic poles for Tarim were reported by *Bai et al.* [1987], but none of these results were substantiated by field tests such as fold, contact or conglomerate tests. A reliable apparent polar wander path for a block as large as Tarim requires paleomagnetic results from many widely distributed sections, and thus we began a comprehensive paleomagnetic sampling program in Tarim to measure the displacement of Tarim with respect to the Siberian, North China and South China Blocks (Figure 1). During the summer of 1986, a paleomagnetic study of the Tarim Block was undertaken jointly by members of the Xinjiang Geology Research Institute, the Beijing Graduate School of China University of Geosciences, Chinese Academy of Geological Sciences and the Geophysics Department at Stanford University. Here, we report Devonian paleomagnetic data as one part of our investigation.

Geological Setting

The Keping-Aksu area lies on the northwestern rim of the Tarim Block (Figures 1 and 2), where the Paleozoic sequences are essentially continuous with the exception of a few disconformities. In a simplified interpretation, the region has been affected by two major phases of tectonic activity since Devonian times. The first is of Late Paleozoic age and may be related to the collision of the Tarim Block with the Tian Shan arc. It is recorded by a disconformity between the Lower Permian Balikelike Formation of marine/shallow-marine facies and the overlying Kalundal Formation of continental facies. The event is also reflected by the change in sandstone petrofacies at the Carboniferous/Permian boundary, from craton-derived to lithic-rich orogen-derived [*Carroll et al.*, 1989]. The emplacement of Permian dikes and basalts in the Keping-Aksu area represent a post-collision extensional event.

The more recent phase of tectonic activity is related to the collision of the Indian plate with the Eurasian plate, an event which began during the Tertiary and continues today [*Xinjiang Stratigraphic Compiling Group*, 1981]. This younger compressional activity is characterized by tilting, folding and thrust faulting around the margins of the Tarim Block [*Xinjiang*

[1] Department of Geophysics, Stanford University, Stanford, California.
[2] Deceased on January 27, 1987.
[3] Geology Research Institute, Geology Bureau of Xinjiang, Urumqi, China.
[4] Geology Institute, Chinese Academy of Geological Sciences, Beijing, China.
[5] Beijing Graduate School, China University of Geosciences, Beijing, China

Copyright 1990 by the American Geophysical Union.

Paper number 90JB01373.
0148-0227/90/90JB-01373 $05.00

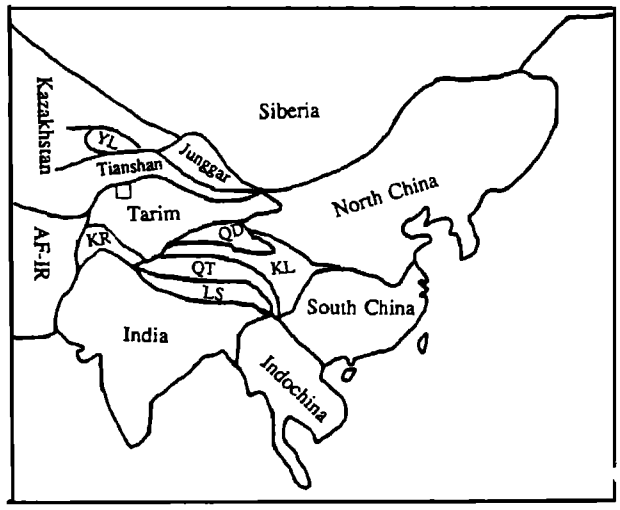


Fig. 1. Generalized tectonic map of Asia. Box is the location of the Devonian paleomagnetic sampling area in the Tarim Block. Abbreviations are AF-IR, Afghan-Iran; YI, Yili; KR, Karakoram; QD, Qaidam; KL, Kunlun; QT, Qiangtang; LS, Lhasa.

Stratigraphic Table Compiling Group, 1981; *Molnar and Tapponier*, 1975]. The most obvious structural features in the Keping-Aksu sampling area are multiple fault blocks which have detached Sinian (latest Precambrian) to Paleozoic stratigraphic packages from the basement along a Cambrian gypsum layer, thrusting them toward the Tarim basin. Relative rotations may have occurred between different fault blocks in the detachment process, and therefore our sampling strategy was to sample rocks from multiple fault blocks to detect possible local rotations.

Samples of Devonian red sandstones were taken at three different localities: Subasi, Acha and Yingan (Figures 2 and 3). The maximum thickness of the Devonian section is 1998 m at Subasi (Figures 2 and 3), where a reference stratigraphic profile at a scale of 1:5000 was surveyed. We then correlated the stratigraphic sections at the other two localities with the Subasi profile using lithologic and magnetostratigraphic information.

The age of the upper part of the Devonian sequence is not well known at any of the sampling localities. At Subasi, the contact between the Devonian red sandstones and overlying Late Carboniferous limestones (the Kangkeling Formation) is disconformable. However, about 50 km north of Subasi, the

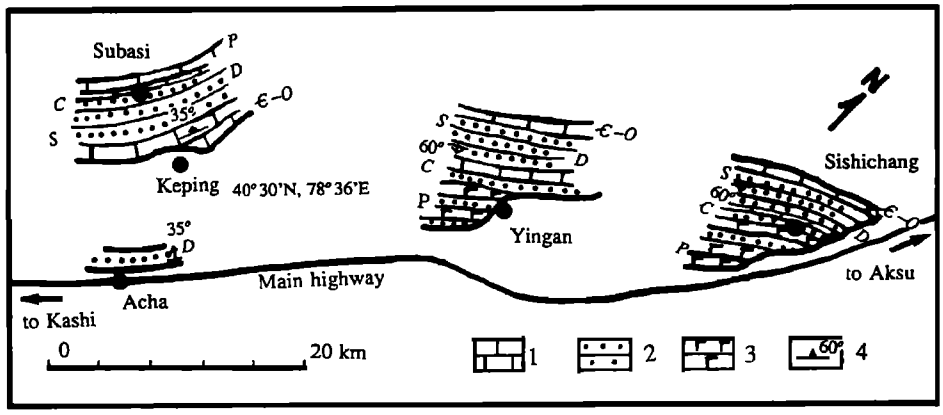


Fig. 2. Sketch of the Devonian paleomagnetic sampling localities in the Keping-Aksu area, Xinjiang, China (a map view). The key shows 1, limestone; 2, sandstone; 3, basalt; 4, dip vector and dip of strata.

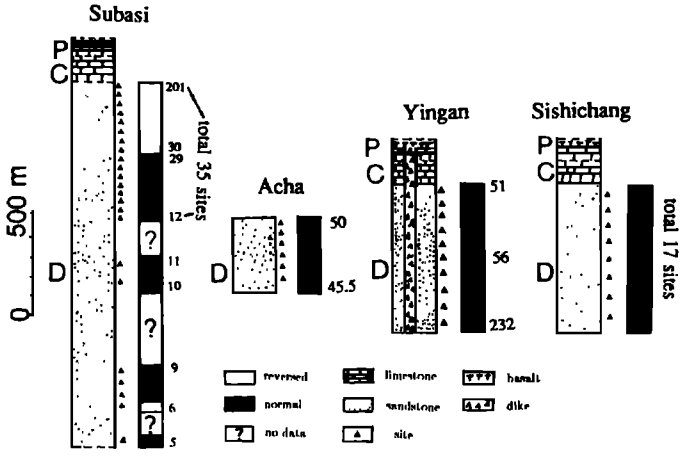


Fig. 3. Stratigraphic and magnetostratigraphic sections at Subasi, Acha, Yingan and Sishichang: D, Devonian; C, Carboniferous; P, Permian. The results from the Sishichang section are based on data of *Bai et al.* [1987].

21562202b, 1990, B12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB12p19185 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

same Devonian red sandstone formation is capped by a conglomerate layer and is overlain by limestones and sandstones of the Lower Carboniferous Lugutubularke Formation. The Lugutubularke Formation contains the coral *Kueichouphyllum sp., Dibunophyllum sp., Arachnolasma sp.;* and the brachiopoda *Gigantoproductus sp.* Subdivision of the Devonian sequence has been accomplished primarily according to sandstone petrology. Fossils in strata near the base of the Devonian sequences include *Edmondia sp.*, the brachiopoda *Barroisella sp., Lingula sp.*, the gastropoda *Naticopsis sp., Euomphalus sp.* and the plant *Lepidodendropsis sp.* [*Xinjiang Stratigraphic Compiling Group*, 1981]. Regionally, a disconformity marks the boundary between the Devonian red sandstones and underlying Lower Silurian sandstones and shales which contain trilobites, graptolites, corals, brachiopoda and gastropoda.

At Subasi, the entire 1998 m Devonian sequence is ideal for paleomagnetic study, with the exception of the lower to middle part of the section where the strata are too fissile to be sampled. A total of 42 sites were located in this section, with sampling interval between adjacent sites ranging from 30 to 50 m in stratigraphic position. Six individually oriented core samples were generally field drilled at each site. Acha (Figure 2), where six sites were collected, is about 20 km southeast of Subasi. Only a fraction of the Devonian strata are exposed here; the missing beds have been eroded or are covered.

Yingan (Figure 2) is about 60 km east of Subasi. The Devonian stratigraphy here is similar to that at Subasi, but the lower part of the section is covered. The Yingan section is structurally complex compared with the very simple structure at Subasi. A northwesterly fault with approximately 50 m of displacement cut the Devonian section before a set of Permian dikes were intruded. The dip direction of the Devonian strata at Yingan is almost opposite to that at both Subasi and Acha, providing the basis for a tilt test. As described below, the tilt test was inconclusive because the magnetization directions are approximately parallel to the rotation poles and thus little directional change is produced upon restoration of bedding planes to paleohorizonal. Fortunately, two Permian dikes cut the Devonian strata at Yingan, allowing a baked contact test. At Yingan, only the upper part of the Devonian section was sampled. A total of 14 sites were collected, of which six sites were taken from the dikes and the baked zones. The Devonian beds sampled by *Bai et al.* [1987] are from Sishichang, which lies about 20 km east of Yingan. Their samples span a stratigraphic interval equivalent to only the middle and upper parts of the Subasi section.

PALEOMAGNETISM

Field-oriented samples were collected using portable drilling equipment and oriented with magnetic and Sun compass (80% of the samples). When Sun azimuths were not available, local magnetic field gradients at sites were verified in the field by reference to a distant point. Little or no difference was noted between magnetic and Sun azimuths , as expected for weakly magnetized sandstones. Magnetic measurements and demagnetization were conducted at Stanford University using an automated cryogenic magnetometer with integrated alternating field (AF) demagnetization coils, and a Schonstedt TSD-1 oven. After thermal demagnetization, a 5 mT peak AF field was applied to each specimen within the field-free space of the magnetometer to reduce any viscous component acquired while being transferred from the oven to the magnetometer. High coercivities were indicated by pilot AF demagnetization experiments, and thus thermal demagnetization was chosen as the optimal cleaning technique. For each sample generally only one specimen was used for measurement and thermal demagnetization. Demagnetization results from each specimen were plotted on orthogonal vector diagrams for inspection [*Zijderveld*, 1967]. Principal component analysis [*Kirschvink*, 1980] was applied to the set of partial remanent magnetization vectors for each specimen to obtain both a characteristic direction and in some cases an overprint with lower unblocking temperature. Characteristic magnetizations of specimens were averaged to obtain site means according to *Fisher's* [1953] method.

PALEOMAGNETIC RESULTS, SUBASI SECTION

Characteristic magnetizations can be isolated in nearly all the Subasi sites. These magnetizations have shallow inclination and easterly or westerly declination. We believe that they represent normal and reversed Devonian magnetizations (we take easterly direction as normal, see Discussion of Paleomagnetic Results). The within-site dispersion varies from site to site. This variation is believed to reflect differences in grain sizes and compositions among sites. For example, the lower part of the section is more heterogeneous in composition and grain size, and therefore the directions within these sites are more scattered as indicated by the k values (the median k value for sites 5-9 is 19, whereas the median k value for the rest of the section is 31).

Magnetic polarity at Subasi can be classified into two intervals. Samples from site 30 and above (Figure 3) yield reversed polarity, which if correlated with Harland *et al.* 's [1982] time scale places site 30 between 367 and 374 Ma. Below site 30, all samples are of normal polarity, with exception of site 6 at the very bottom of the section. The divisions in the Devonian strata, unclear from geological evidence, are rather distinct in terms of magnetic polarity. For simplicity, the whole profile is divided into two parts: a normal interval which represents site 29 and below (Figure 3), and a reversed interval which represents site 30 and above. There are no Permian secondary magnetizations present in the Subasi section in contrast to the Yingan section, (see Paleomagnetic Results, Yingan Section). The viscous component removed in the lower temperature range (usually below 500°C) is a present Earth field (PEF) overprint. Figure 4 shows some typical thermal demagnetization orthogonal diagrams. The characteristic component can be isolated in most, if not all, of the specimens. The directions within and between sites are consistent.

Magnetic Polarity

Reversed (westerly declination) polarity is found in sites from site 30 and above and site 6. The stratigraphic distance between site 29 (normal) and site 30 (reversed) is about 30 m, and we have no record of transitional directions. The mean declination of site 29 is almost 180° away from that of site 30, suggesting that the reversal is completed in the interval between the two sites. The magnetizations at site 5 and site 6 are less well constrained compared with other sites, probably due to coarser grain sizes in this part of the section. Within the reversed

21562202b, 1990, B12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB12p19185 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

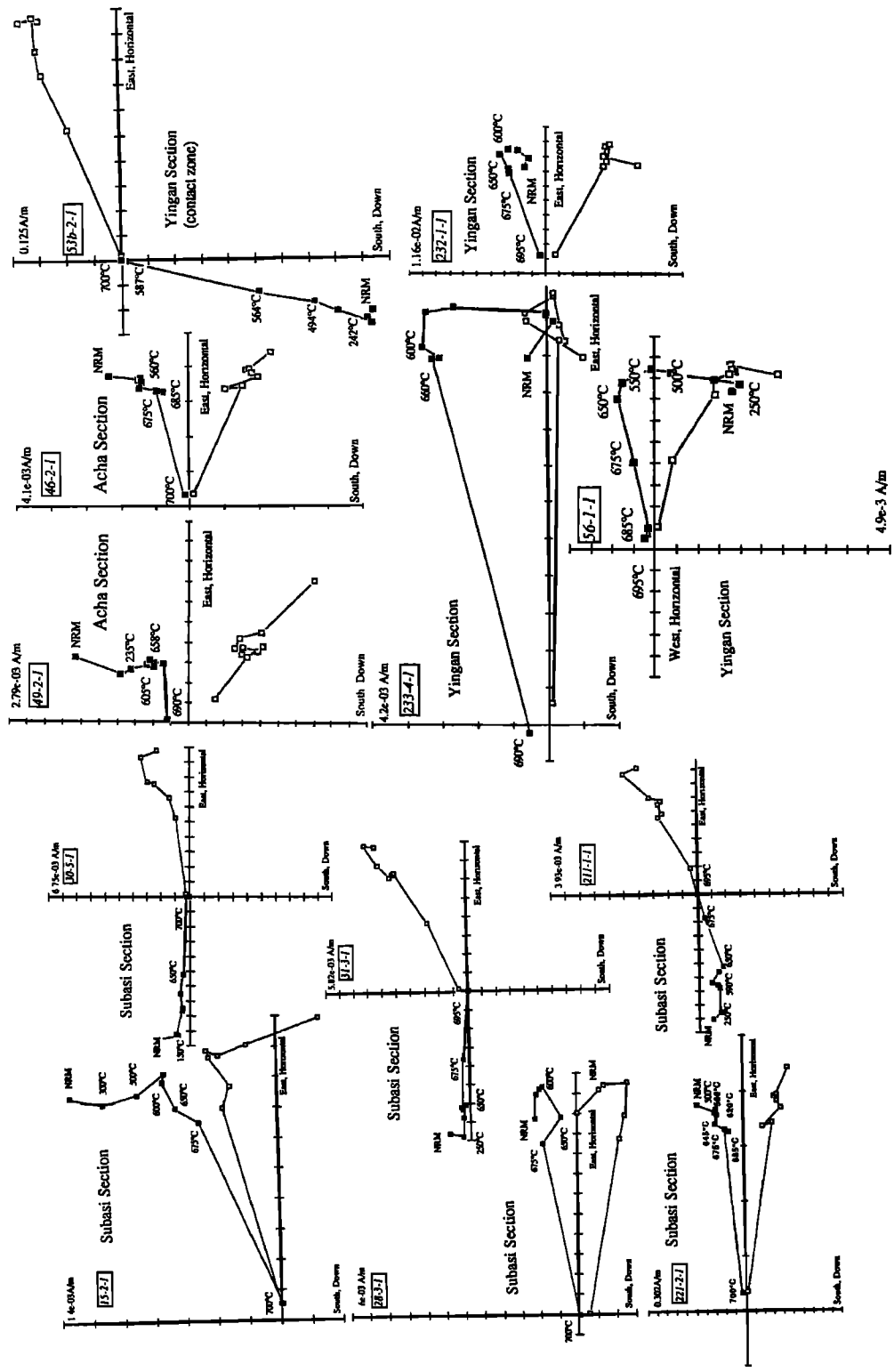


Fig. 4. Orthogonal vector plots of thermal demagnetization of Devonian red bed specimens in this study (geographic coordinates). Solid square symbol is projection on horizontal plane; open square symbol is true inclination.

21562202b, 1990, B12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB12p19185 by Curtin University Library, Wiley Online Library on [24/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

interval, one sample (of six) from site 206 shows normal polarity. Since only one sample shows normal polarity and the area of the sampling site is of 10 x 10 m^2, we think that it probably represents an orientation error rather than a true field reversal. The mean directions of each site at Subasi are plotted in Figure 5 and are listed in Table 1.

The overall normal direction is about 180° away from the reversed direction in declination. However, inclinations are not exactly opposite (Figure 6). This asymmetry could be due to inclination error, incomplete demagnetization or apparent polar wander which are discussed below.

Inclination Error

Inclination error has been a classic explanation for anomalous shallow inclinations observed in sedimentary rocks [*King*, 1955]. Since a chemical remanent magnetization is believed to be the main remanent magnetization of red beds [*Nagata*, 1961], an inclination error may not exist. Even if an inclination error exists, the observed normal and reversed inclinations should still be symmetrical about the horizontal plane which is not the case here. Thus inclination error cannot explain the asymmetry in normal and reversed inclination.

Incomplete Demagnetization

Suppose a secondary magnetization with negative (upward) inclination is superposed on the characteristic component, yet demagnetization does not remove this component. In this case the normal component will be shallower than the mean value, while the reversed component will be steeper than the mean value. However, there is no evidence on orthogonal diagrams to substantiate this vertical component, and this is an unsatisfactory explanation (Figure 4).

Apparent Polar Wander Recorded in the Subasi Section

The Tarim Block moved from south to north from the Devonian to the Carboniferous based on paleomagnetic data (*Zhai et al.* [1988] and this study). The paleolatitude of the reversed interval, which is higher in stratigraphic position, is indeed further north than that of normal interval. Thus the difference between the inclinations of the normal and the reversed intervals may be due to apparent polar wander produced by northward movement during Devonian time. If this northward translation exists, we might see a systematic increase in inclination with increasing stratigraphic position. To test this hypothesis, we plot inclination versus depth from the normal-reversed polarity boundary, somewhere between sites 29 and 30 (Figures 3 and 6). The results show an 8.3° ± 4.7° increase in inclination per 1000 m increase in stratigraphic position (N = 40). According to this calculation, there is a northward trend within the Subasi section. The maximum increase in inclination (taking the upper limit of the slope) for the whole Devonian section at Subasi is 24°, about the difference between the mean inclinations of the reversed and normal polarity intervals. Therefore apparent polar wander could account for the systematic increase in inclination for this portion of the Devonian. If we assume that the section represents 40 Ma (Devonian is from 360 to 408 Ma [*Harland et al.*, 1982]), this corresponds to a poleward speed of 47 ± 19 km/Ma (4.7 cm/yr).

PALEOMAGNETIC RESULTS, ACHA SECTION

Examples of orthogonal plots of samples from Acha are plotted in Figure 4. Directional data are displayed in Figure 5 and listed in Table 2. All site mean directions at Acha are of a single polarity, but the distribution of directions is elongated.

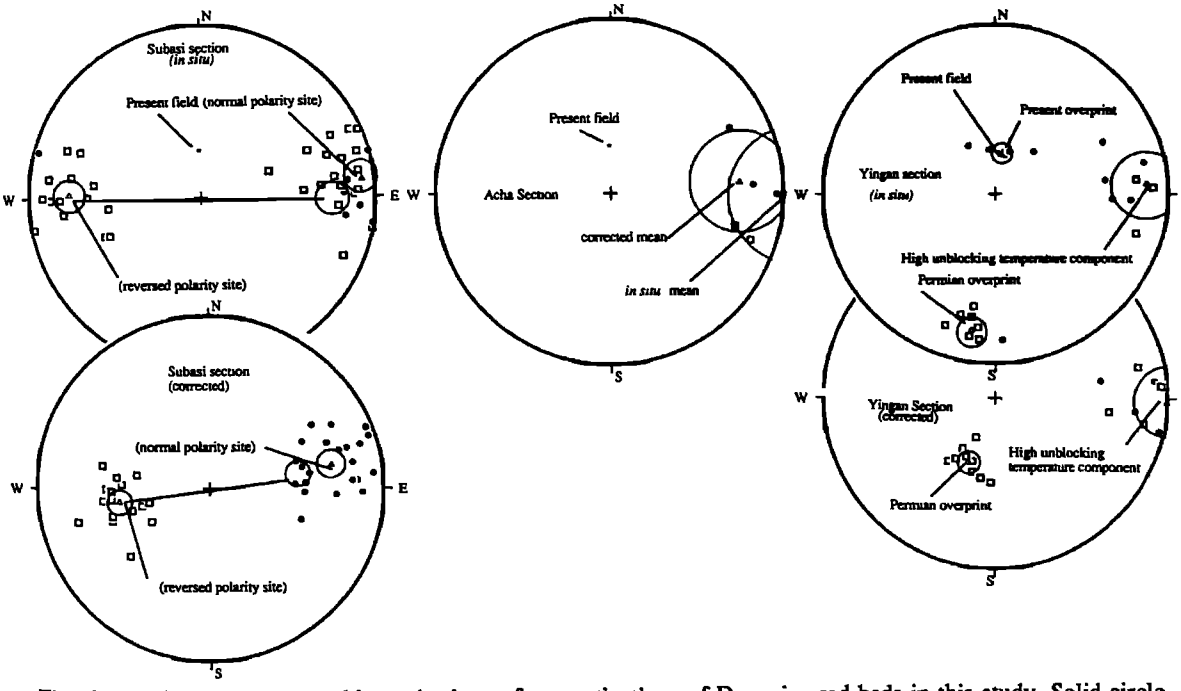


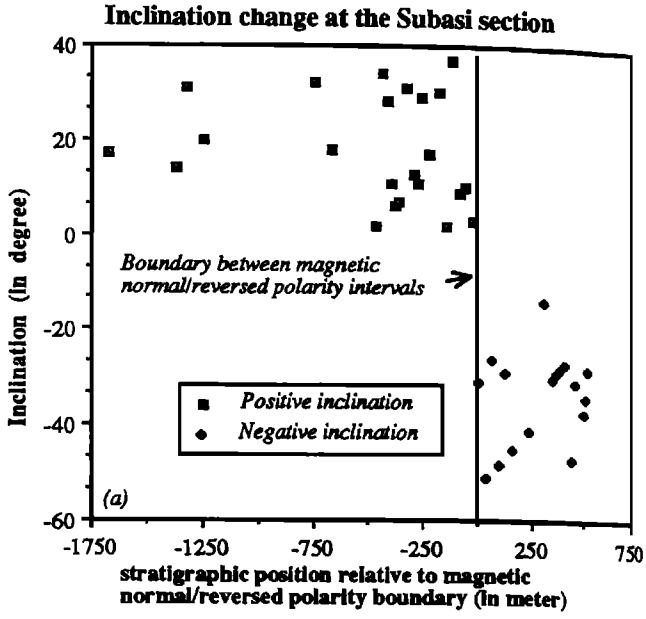
Fig. 5. Equal angle stereographic projections of magnetizations of Devonian red beds in this study. Solid circle (open box) is a site mean direction plotted on the lower (upper) hemisphere. Solid (open) triangle is the grand mean direction plotted on lower (upper) hemisphere. For the Subasi section, the grand mean directions of reversed polarity sites are inverted into normal polarity and are also plotted, connected with lines. The α_{95} for the mean directions are plotted as small circles about the means. Permian overprints (by dikes) in the Yingan section are also plotted.

21562202b, 1990, B12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB12p19185 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Devonian Paleomagnetic Data of Subasi, Tarim

Site	N	D_b	I_b	D_a	I_a	k	a_{95}	Polarity
215	6	286	2	282	-26	13	16	R
214	5	263	-34	239	-48	37	13	R
213	6	263	-12	251	-29	140	5	R
212	6	291	-17	281	-45	35	10	R
211	6	270	-24	253	-41	45	9	R
210	6	259	-1	255	-14	74	8	R
209	6	263	-13	253	-30	319	3	R
208	4	248	-29	269	-29	11	29	R
207	5	269	-8	262	-28	34	11	R
206	6	270	-5	263	-27	22	15	M
205	6	278	-27	255	-47	30	12	R
204	6	269	-12	263	-31	28	13	R
203	6	279	-16	272	-37	44	10	R
202	5	290	-12	277	-34	25	16	R
201	6	247	-31	229	-28	160	5	R
31	5	277	-27	256	-51	38	10	R
30	6	277	-7	268	-31	54	8	R
29	6	82	-14	85	3	27	13	N
28	6	81	-5	80	10	26	14	N
27	6	86	-12	87	9	21	15	N
26	6	98	10	88	37	76	8	N
25	6	67	-22	72	2	83	7	N
24	5	84	9	74	30	29	14	N
23	6	75	0	72	11	20	15	N
22	6	94	5	87	31	33	12	N
21	4	70	-44	87	-21	24	19	N
20	6	72	-15	74	6	14	19	N
19	5	85	-20	87	11	14	21	N
18	5	66	-4	63	13	75	9	N
17	4	94	-13	93	17	55	12	N
16	6	88	-6	82	29	88	7	N
15	5	87	-28	92	7	24	16	N
14	5	77	9	64	28	72	9	N
13	6	89	11	73	34	31	12	N
12	6	65	-11	69	2	18	16	N
11	6	74	-4	69	18	18	13	N
10	2	101	0	93	32	-	-	N
9	4	67	-1	59	20	12	28	N
8	4	113	-7	109	31	84	10	N
7	6	75	-10	74	14	117	6	N
6	5	258	-31	220	-49	19	18	R
5	4	90	7	78	32	17	23	N
$Mean_b$ [a]	24+	83	-5.4	-	-	24.9	6.2	N
$Mean_a$ [a]	24+	-	-	79.1	18.9	25.8	6.1	N
$Mean_b$ [r]	17+	271	-16	-	-	25	7.3	R
$Mean_a$ [r]	17+	-	-	261	-35	29	6.7	R
$Mean_b$	42+	86	3	-	-	14.8	5.9	-
$Mean_a$	42+	-	-	79	25	18	5.3	-

D_b (D_a) and I_b (I_a) are site mean directions before (after) tilt corrections; N is number of samples or sites (+). $Mean_b$ [a] ($Mean_a$ [a]) is mean direction of all sites of normal polarity before (after) tilt corrections; $Mean_b$ [r] ($Mean_a$ [r]) is mean direction of all (except for site 6) sites of reversed polarity before (after) tilt corrections. $Mean_b$ ($Mean_a$) is mean direction of all sites before (after) tilt corrections.



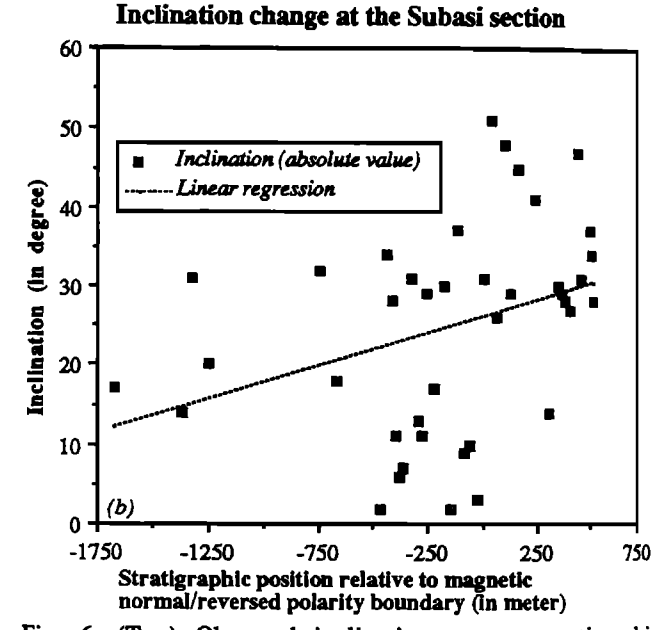


Fig. 6. (Top) Observed inclinations versus stratigraphic position from the normal-reversed polarity boundary at the Subasi section; (Bottom) Linear regression of the inclination with respect to stratigraphic position. The slope of the regression line is 8.48° ± 3.7°/km.

The between-site dispersion is primarily in declination, which could result either from incomplete removal of secondary magnetizations, or from insufficient sampling of the geomagnetic field. The average within-site k value is about twice the between-site k value, indicating that some secular variation of the geomagnetic field has been averaged. Permian igneous rocks are not exposed at this locality, and there is no Permian magnetic overprint present, in contrast to the Yingan section (see Paleomagnetic Results, Yingan Section). However, present field viscous overprinting persists to 550°C during thermal demagnetization (Figure 4) for several samples from this locality, and the characteristic magnetization was based on final high temperature steps (> 550°C) for these samples. Because the declinations of all sites are approximately east-directed which we believe to be the normal polarity for the Devonian paleomagnetic directions from Tarim, combined with field lithological observation, we think that the stratigraphic position of this section corresponds to the middle and lower parts of the Subasi section (see Paleomagnetic Results, Subasi Section and Figure 3).

PALEOMAGNETIC RESULTS, YINGAN SECTION

Progressive thermal demagnetization of specimens from the Yingan section suggests the presence of three magnetizations: a PEF secondary magnetization, a Permian secondary magnetization and a primary (?) Devonian magnetization. As an example, these three components can be isolated from specimen 56-1-1 (Figures 4 and 7).

21562202b, 1990, B12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB12p19185 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Devonian Paleomagnetic Data of Acha, Tarim

Site	N	D_b	I_b	D_a	I_a	k	a_{95}	Polarity
45.5	6	46	-11	54	-15	34	12	N
46	6	62	14	53	14	43	10	N
47	4	109	-9	107	17	20	21	N
48	4	91	2	83	23	15	24	N
49	5	87	11	76	25	32	12	N
50	5	105	-17	106	11	39	12	N
Mean(b.c.)	5*	91	0	-	-	13	22	N
Mean(a.c.)	5*	-	-	85	19	13	22	N

D_b(D_a) and I_b(I_a) are site mean directions before (after) tilt corrections; N is number of samples or sites (*). Site 45.5' is excluded from average.

PEF Overprint

PEF overprints were observed at all sites, and were isolated by inspection of subtracted vectors or a least squares fit in the temperature range between natural remanent magnetization (NRM) and 300°C. For 9 sites we obtained an *in situ* mean PEF direction of $D = 10°$, $I = 63°$, $k = 62$, $a_{95} = 7°$, indistinguishable from the present Earth magnetic field direction ($D = 357°$, $I = 59°$) or geomagnetic dipole field direction of the locality ($D = 0°$, $I = 60°$).

Permian Overprint

Permian paleomagnetic results from northwestern Tarim have been reported elsewhere [*McFadden et al.*, 1988; *Li et al.*, 1988; *Sharps et al.*, 1989]. A presumed Permian overprint is present at Yingan. Specimens located within the baked zones in contact with Permian dikes (within about 10 m of the contact) are dominated by magnetizations which have directions identical to those of the dikes, have an unblocking temperature characteristic of magnetite, and can be isolated by alternating field (AF) demagnetization (Figure 4). A Devonian (?) characteristic component cannot usually be recovered from the majority of specimens within the baked zone. However, there are some exceptions (Figure 8 and Table 3).

Sites more than 10 m from the contact yield primarily a characteristic magnetization with easterly declination and shallow inclination, but they are still not univectorial. A Permian overprint is clearly present during thermal demagnetization to 580°C (Figures 4, 5 and 7), above which the Devonian characteristic component dominates, but the Permian overprint can be seen during demagnetization. The component intensity spectra of one three-component specimen (Figure 7) from Yingan show that: (1) the PEF overprint component is removed below 250°C; (2) the Permian overprint component is removed below 600°C, suggesting that magnetite is likely to be the main magnetic carrier and (3) the magnetic carrier of the highest unblocking temperature characteristic component is hematite. The component intensity spectrum is the relative contribution of each component isolated by the principal component analysis to the total vector during demagnetization. It is calculated by inverting a matrix of the stepwise demagnetization vectors to solve for the component intensity matrix [*Li*, 1988] (see Appendix).

Devonian Direction

Apart from sites 51 and 53 which are near the dike contact and are dominated by a Permian overprint, all other sites at Yingan yield a high unblocking temperature magnetization similar to those obtained at Subasi and Acha. This magnetization has an unblocking temperature characteristic of hematite, as opposed to the lower unblocking temperatures seen in the Permian overprint. We believe this component is a Devonian characteristic magnetization which is best characterized in specimens such as 232-1-1 and 233-4-1 (Figure 4). However, the component may not represent a true Devonian magnetization because it has been affected by the Permian overprint and thermal demagnetization may not have completely removed the Permian overprint. The problem will become obvious later when a comparison is made with the sections at Subasi and Acha. Site and section mean directions of each components are shown in Figure 5 and are listed in Table 3.

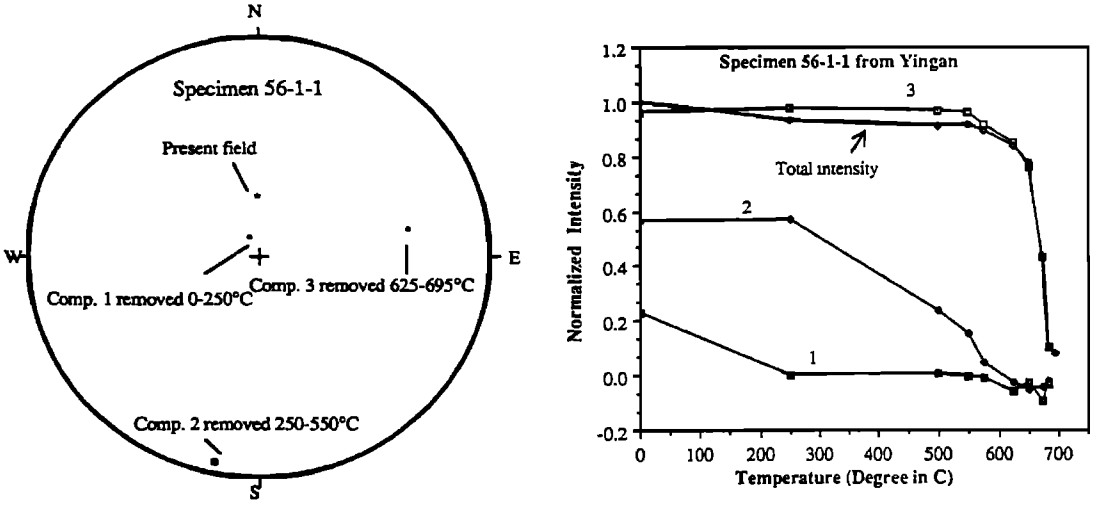


Fig. 7. (Left) Equal area stereographic projection of the components isolated by principal component analysis for specimen 56-1-1 from the Yingan section (all in geographic coordinates). (Right) Relative intensity spectra of components isolated from the same specimen. The presumed PEF overprint (component 1 as in the diagram on the left) and the presumed Permian overprint (component 2 as in the diagram on the left) are removed during low-temperature range.

21562202b, 1990, B12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB12p19185 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

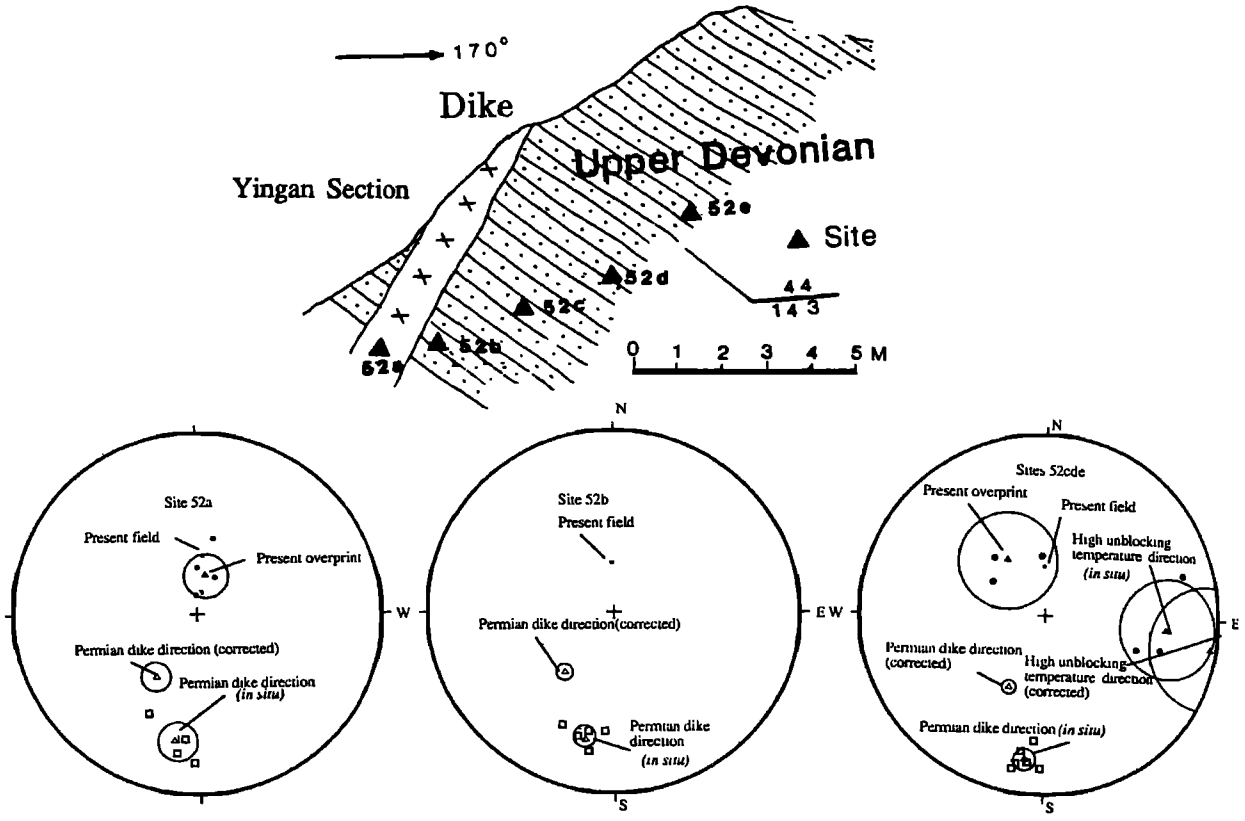


Fig. 8. Sketch of the contact between the Devonian red beds and a Permian dike at Yingan along with equal angle stereographic projections of the mean directions of each site. Symbols used are the same as in Figure 3.

Baked Contact Test

Easterly and shallow magnetization directions are found in samples far removed from the dike contacts. Since these directions are strikingly different from the directions seen in the Permian dikes and their baked contacts, we infer that the characteristic magnetization in the Devonian sandstones predates the Permian dikes (Figures 4 and 8). Since the magnetizations of the dikes pass a fold test and were probably acquired during the Kiaman reversed interval *[Li et al.*, 1988], we conclude that the high unblocking temperature magnetization in the Devonian strata is of pre-Kiaman age (250-320 Ma, [*Harland et al.*, 1982]).

TABLE 3. Devonian Paleomagnetic Data of Yingan, Tarim

Site	N	D_b	I_b	D_a	I_a	k	a_{95}	Polarity
51	6	200	-11	215	-41	38	11	-
52	3+	95	23	100	-7	14	22	N
52.5	5	95	15	102	1	6	26	N
53	3+	190	-22	211	-52	214	6	-
54	5	81	26	96	11	17	15	N
56	6	80	9	84	4	29	11	N
232	6	67	22	81	26	118	5	N
233	6	84	-11	78	-9	19	16	N
234	2	88	-5	86	-2	-	-	N
235	3	106	-9	97	-22	76	14	N
PEF	9+	10	63	-	-	62	7	N
P^{o_b}	9+	189	-11	-	-	68	6	R
P^{o_a}	9+	-	-	201	-46	55	7	R
DEV_b	8+	87	9	-	-	18	13	N
DEV_a	8+	-	-	91	0	23	12	R

D_b (D_a) and I_b (I_a) are mean directions before (after) tilt corrections; sites 52 and 53 which each consists of three subsites were collected from the Permian dikes and baked contact zones; N is number of samples or sites (+). PEF is the present Earth field overprint direction; P^{o_b} (P^{o_b}) is a Permian overprint direction before (after) tilt correction; DEV_b (DEV_a) is the site-mean Devonian direction before (after) tilt correction (sites 52 and 53 are Permian dikes and baked contact zones, excluded in average).

MAGNETIC CARRIERS AND ROCK MAGNETISM

Bulk Magnetic Susceptibility

To compare carriers of remanence at the three sampling localities and to monitor mineralogical changes during thermal demagnetization, bulk magnetic susceptibility measurements were made for several groups of red sandstone specimens after each heating step.

Specimens from Subasi site 31 (Figure 9) display only minor changes during heating to 695°C. A slight increase in susceptibility was noted above 550°C for specimens from site 31, which is probably due to a chemical change during the heating. The initial susceptibilities of samples from Yingan site 56, where Permian basalts and dikes are exposed, were approximately twice those of samples from site 31 at Subasi (Figure 9). Upon heating, these higher susceptibilities decreased with increasing temperature to 500°C, approaching similar values to those displayed in sites far removed from the Permian dikes. After 550°C a slight increase in susceptibility

21562202b, 1990, B12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB12p19185 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

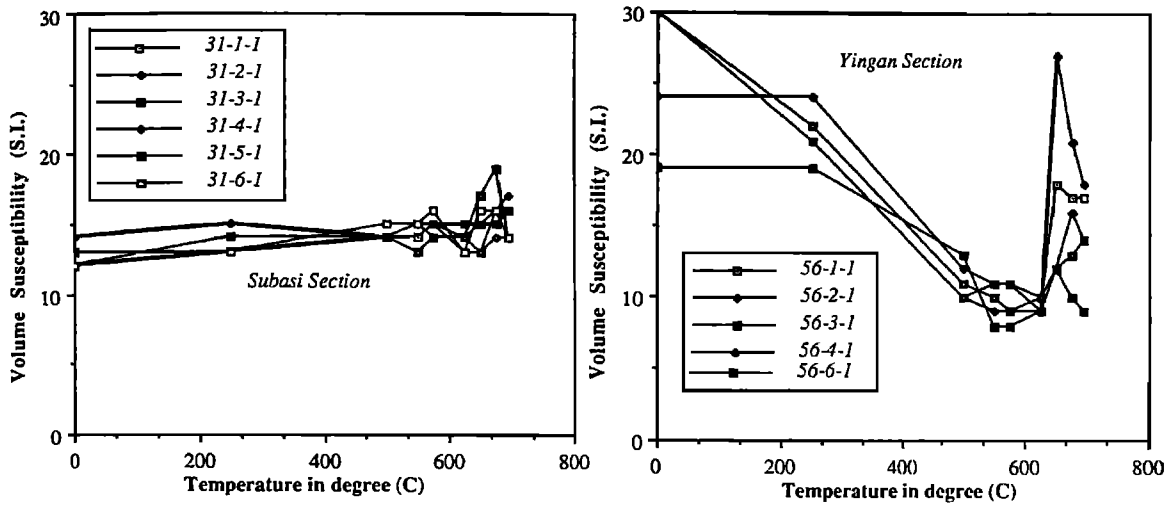


Fig. 9. Susceptibility versus temperature for specimens of site 31 of the Subasi section, and site 56 of the Yingan section.

with increasing temperature was observed, as for samples of site 31 at Subasi. The decrease in susceptibility observed between 200°C and 500°C is probably related to the multicomponent magnetizations present at Yingan, where the presumed primary magnetization of the Devonian sandstones was partially overprinted during Permian times. The intensity changes observed during thermal demagnetization and saturation isothermal remanent magnetization (SIRM) acquisition experiments suggest that magnetite coexists with hematite in the specimens from the Yingan section (Figures 7 and 10). The magnetite is probably a product of hydrothermal circulation linked to Permian igneous activity and is responsible for the overprint. Higher initial susceptibilities in the Yingan samples are due to the presence of secondary magnetite. Upon heating, the magnetite oxidized to hematite, and produced a decrease in susceptibility. The susceptibility changes mirror the directional changes seen as the Permian overprint is progressively removed.

SIRM Acquisition Experiments

A few red sandstone specimens which were not subjected to thermal demagnetization were selected for SIRM acquisition experiments, as shown in Figure 10. The SIRM curves indicate that the primary magnetic carrier in these specimens is hematite. Relatively rapid increases in intensity of SIRM for specimens 51-1-2 and 54-5-1 in lower magnetic fields applied may be attributed to minor amount of magnetite resulted from hydrothermal contamination, because these two specimens are from the Yingan section where Permian basalts and dikes are exposed.

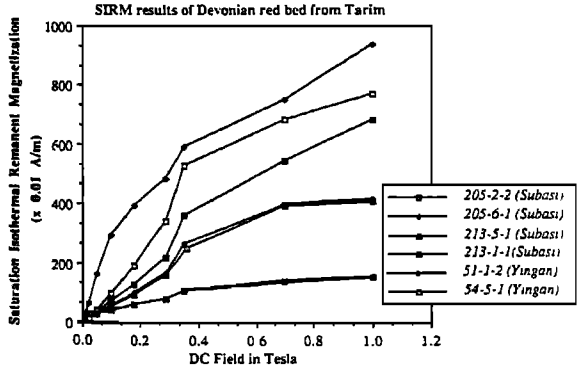


Fig. 10. Plot of results of SIRM acquisition experiments for six Devonian red bed specimens. Specimens 51-1-2, 54-5-1 are from the Yingan section; specimens 205-6-1, 205-2-2, 213-1-1 and 213-5-1 are from the Subasi section.

Discussion of Paleomagnetic Results

Characteristic magnetic directions based mainly on the results of thermal demagnetization were isolated from 59 sites. The characteristic directions from the Devonian strata are significantly different from the PEF direction and any known later Carboniferous, Permian, Jurassic, and Cretaceous directions. The observation of apparent polar wander from the Devonian to the Permian within the same continuous stratigraphic section at Subasi [*Sharps et al.*, 1989; *Zhai et al*, 1988], tends to rule out the possibility of regional post-Permian remagnetization. The characteristic magnetic directions from the Subasi and Acha sections are therefore believed to be pre-Carboniferous in age.

According to the observed magnetic polarities, we believe that the stratigraphic positions of the Acha and Yingan sections are below site 29 but above site 10 of the Subasi section. The Devonian Sishichang section sampled by *Bai et al.* [1987] is about 20 km east of Yingan (Figure 2). The polarity of all sites at the Sishichang section is normal (east-pointing declination), the same as those of the Yingan and Acha sections. Based on field observations and magnetic polarity correlation, we believe that the stratigraphic position of the Sishichang section is also above site 10 and below site 29 of the Subasi section. Thus, apart from the Subasi section, all other sections have approximately the same stratigraphic position, within the interval between sites 10 and 29 of the Subasi section (Figure 3).

The Devonian magnetizations isolated from the Subasi section do not differ significantly from those of the Acha section, suggesting a negligible relative rotation between these two localities (Table 4). However, the Devonian pole from the Subasi section is statistically different from both

21562202b, 1990, B12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB12p19185 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4. Devonian Paleomagnetic Data from the Tarim Block, China

Locality	N	D_b	I_b	D_a	I_a	k	a_{95}	Polarity
Subasi(N)$_b$	24	83	-5	-	-	25	6.2	N
Subasi(N)$_a$	24	-	-	79	19	26	6.1	N
Subasi(R)$_b$	17	271	-16	-	-	25	7.3	R
Subasi(R)$_a$	17	-	-	261	-35	29	6.7	R
Subasi$_b$	42	86	3	-	-	14.8	5.9	-
Subasi$_a$	42	-	-	79	25	18	5.3	-
Yingan$_b$	8	87	9	-	-	18	13	N
Yingan$_a$	8	-	-	91	0	23	12	N
Acha$_b$	5	91	0	-	-	13	22	N
Acha$_a$	5	-	-	85	19	13	22	N

Subasi$_b$ (Subasi$_a$) refers to the site-mean direction at the Subasi section before (after) tilt correction; Subasi(N)$_b$ (Subasi(N)$_a$) refers to site-mean direction of all Subasi normal-polarity sites before (after) tilt correction; Subasi(R)$_b$ (Subasi(R)$_a$) refers to site-mean direction of all Subasi reversed polarity sites before (after) tilt correction; N is number of sites.

those of the Yingan (this study) and Sishichang sections [*Bai et al.*, 1987] (Figure 11). The difference may be ascribed to one or more of the following phenomena:

1. The Yingan and the Sishichang sections may have been rotated about a vertical axis with respect to the Subasi and Acha sections. The paleomagnetic directions of the Permian dikes at Yingan are indistinguishable from Permian direction from Bachu, some 250 km west of Yingan [*Li et al.*, 1988], and so are the Carboniferous directions from both Yingan and Subasi sections [*Zhai et al.*, 1988]. If the differences are due to block rotations only, then the colatitudes for each location sampled should all be the same. However, magnetic inclinations from the Subasi, Acha, Yingan and Sishichang Devonian sections are also different.

2. Apparent polar wander (APW) during the Devonian. The amount of APW within the Subasi section, the longest Devonian stratigraphic section studied, is not large enough to account for differences between the directions from Subasi, Acha, Yingan and Sishichang. Based on paleomagnetic polarity correlation and geological field observations, we think that the lowest level of the Devonian strata sampled in this region is at Subasi. If any apparent polar wander occurred during the Devonian, poles from Sishichang and Yingan should lie on a small circle segment with the Subasi and Acha Devonian pole and the Permian pole defining the ends. Instead, the Devonian poles from Sishichang and Yingan do not lie on the segment, but in fact lie on the outside of the segment, toward an opposite direction (Figure 11).

3. Overprinting by the Permian igneous activity. Apart from the magnetic characteristics which were discussed earlier, the difference in magnetic directions from the Devonian sections correlates well with the presence of the Permian igneous activity. Both the Subasi and Acha sections are devoid of any Permian basalts and dikes, whereas both the Yingan (this study) and Sishichang sections [*Bai et al.*, 1987] were intruded by Permian basalts and dikes. The overprint produced by the Permian dikes and basalts is recognized in the Yingan Devonian demagnetization diagrams (Figures 4, 7 and 8). Although a Devonian characteristic magnetization was isolated from the Yingan section, it may be biased toward the directions of the Permian basalts and dikes. If this hypothesis is correct, the contaminated Yingan Devonian pole would lie on a plane defined by the Permian pole (reversed polarity) and the Subasi Devonian pole (normal polarity). Figure 11 shows that the poles from the Sishichang and Yingan sections do lie on a plane defined by the observed Permian pole and the Subasi Devonian pole. The rock magnetic study described above also indicates that the magnetization of the Devonian red sandstones at Yingan are contaminated by the Permian basalts and dikes, possibly through hydrothermal circulation. This explanation is believed to be the most reasonable. A Devonian pole calculated from results of only Subasi and Acha suggests that the high rate of apparent polar wander between the Devonian and Carboniferous reported previously by *Bai et al.* [1987] is an artifact produced by incomplete removal of a Permian overprint. A similar analysis reveals that the Carboniferous pole given by *Bai et al.* [1987] may also be contaminated by a Permian overprint.

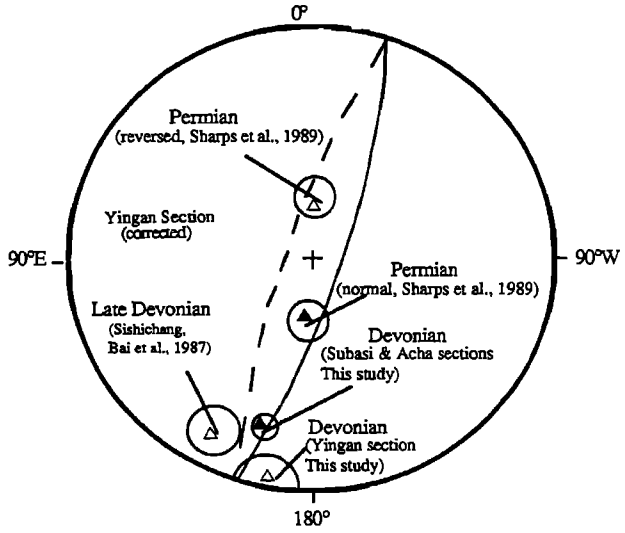


Fig. 11. Equal angle stereographic projections of Devonian, Permian Paleomagnetic poles from Tarim and a great circle drawn on them. Solid (open) triangle is plotted on lower (upper) hemisphere.

Although the high unblocking temperature magnetizations carried by hematite from Yingan are contaminated by a Permian secondary magnetization, we believe the high unblocking temperature characteristic magnetizations from the Subasi and Acha sections are pre-Carboniferous in age. The grand mean directions from 47 sites from the Acha and Subasi sections were first converted into virtual geomagnetic poles and then were averaged according to *Fisher's* [1953] method. The mean pole position is ϕ_p = 169.0°E, λ_p = 5.0°N, K = 25, $A95$ = 4.3° before tilt correction, and ϕ_p = 165.0°E, λ_p = 16.5°N, K = 25 and $A95$ = 4.3° after tilt correction. The inconclusive tilt test is due to the fact that the magnetic directions are roughly aligned with the strike of the strata. Even if the fold test were positive, it would constrain the age of the magnetizations to be only pre-Early Tertiary, as the last stage of folding and tilting in the region is related to the collision between the Indian and Eurasian plates [*Chang et al.* 1986; *Xinjiang Stratigraphic Compiling Group*, 1981].

On the basis of the baked contact test, presence of the dual magnetic polarities within the same continuous stratigraphic section and a reasonable spatial progression of poles from the Devonian to the Permian, we believe that the remanent magnetizations isolated from the Subasi and Acha sections are primary Devonian directions. Without the Early and Middle Carboniferous paleomagnetic poles from Tarim, we cannot

21562202b, 1990, B12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB12p19185 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

determine the absolute polarity of the Devonian magnetic directions. However, we believe the easterly-directed magnetization are normal, because otherwise the Tarim Block would have translated 27° northward and rotated 160° counterclockwise between Devonian and Carboniferous times. Although this may be possible, in the absence of evidence to the contrary, we choose the minimum-displacement alternative.

TECTONIC EVOLUTION OF THE TARIM BLOCK

When the Tarim Block became a tectonic coherent block is unknown at present. Reported Permian paleomagnetic results from the Hetian Region in southwest Tarim, reconciled with the Permian results from the Keping-Aksu region, indicate that Tarim has been a coherent tectonic block since Permian [*Li et al.*, 1989; *Sharps et al.*, 1989]. This argument can be probably extended to Devonian time as Devonian paleomagnetic data from southwestern Tarim also resemble those we report here (Z. Meng, oral communication, 1989). Pre-Devonian paleomagnetic poles are not available from the Tarim Block except for some Sinian (Late Precambrian) data reported from the Wushi region, 50 km north of the Keping and Yingan sections [*Li et al.*, 1990]. The Sinian paleomagnetic data are based on the directions isolated from basalts, limestones and sandstones which show dual polarity. The mean Sinian inclination is 16° (positive or negative). Tarim was probably located in low latitudes during the Early Paleozoic. Our paleomagnetic results support a northward translation and a clockwise rotation for the Tarim Block since the Devonian. Figure 12 shows that the Tarim Block also occupied a low-latitude position during the Devonian.

The Late Carboniferous pole of *Bai et al.* [1987] is contaminated by the Permian igneous activity, but a new Late Carboniferous pole has been reported by *Zhai et al.* [1988]. Significant (15°) northward translation for the Tarim Block occurred between the Devonian and the Late Carboniferous (Figure 12). Northward translation during the Permian was small, indicating that the poleward velocity of the Tarim Block was reduced and a continent-continent or continent-arc collision was probably involved. Between the Devonian and the Early Permian, the Tarim Block moved about 15° northward and rotated 40° clockwise. The collision model will depend on more reliable Early Permian and Late Carboniferous poles from both the Tarim block and the Tian Shan arc, however, one interpretation attaches Tarim to an oceanic plate subducting to the north under the Tian Shan volcanic arc since a subducting oceanic plate often has a high plate speed [*Gordon et al.*, 1984]. The rotation probably resulted from an initial collision between Tarim's eastern end and the Tian Shan arc. Sedimentary facies analysis in the Keping-Aksu area suggests that a paleo-ocean opened toward the northwest during Carboniferous and Permian times [*Zhang et al.*, 1983]. The magnitude of strike-slip motion between the Tarim Block and the Tian Shan since collision is unknown, and the tectonic effects (metamorphism, volcanism) due to the collision may not be stronger in present eastern Tian Shan than western Tian Shan. If the collision did start at the east end, we would expect to observe that the tectonism due to the collision migrated from east to west with decreasing age. Indeed, volcanism and tectonic activity along the Tian Shan's southern margin migrated from east to west according to *Ren et al.*, [1987]. This collision model should not be confused with the general suturing process between the

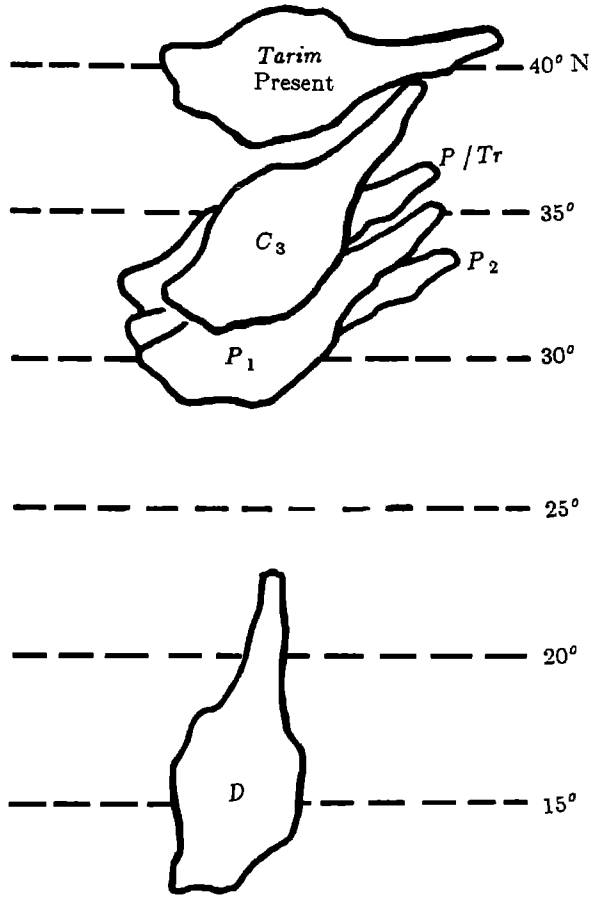


Fig. 12. Northward translation history of the Tarim Block inferred from paleomagnetic data. Age and data source: P/Tr, Permian/Triassic [*McFadden et al.*, 1988]; P_2, Late Permian [*Li et al.*, 1988]; P_1: Early Permian [*Sharps et al.*, 1989; *Bai et al.*, 1987]; C_3: Late Carboniferous [*Zhai et al.*, 1988]; D: Devonian (this study).

Angaran paleoplate (Siberia, Kazakhstan) and Cathaysian paleoplate (North China, Tarim and others) [*Zhang et al.*, 1984], which is characterized by a progressive closure from the west (in the Altay belt) during the Carboniferous, to the east (in the northern Sea of Okhotsk) during the Cretaceous [*Nie et al.*, 1990; *Zhao et al.*, 1990]. A scissor-type collision has also been proposed for between the North China and South China Blocks [*Zhao and Coe*, 1987]. The examples of collision between the Asian blocks indicate that an oblique collision may be common for a continent-continent collision.

Northward translation and relative closure between Early and Late Permian times suggest that the collision between the Tarim Block and the Kazakhstan Block was not finished by the Early Permian [*Sharps et al.*, 1989; *Li et al.*, 1988]. Analogies with the collision of India and Tibet seem apt. Since the Late Permian, Tarim has moved about 10° northward and has rotated clockwise by 20°, placing it in its present position.

DEVONIAN PALEOMAGNETIC DATA FROM OTHER BLOCKS OF ASIA

Although no Devonian paleomagnetic data are available from the North China Block (NCB), three Devonian paleomagnetic results have been reported from the South China Block (SCB). An Early-Middle Devonian paleomagnetic result from red beds

21562202b, 1990, B12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB12p19185 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

in the eastern part of the South China Block is reported by *Tan and Zhai* [1988]. The result is from Nanjing and Wuxi of southern Jiangsu Province which has dual polarity and passes the fold test at the 99% confidence level [Zhushan: $D = 212°$, $I = -14°$, $a_{95} = 7°$; Jikang: $D = 354°$, $I = 12°$, $a_{95} = 7.3°$ and $D = 174°$, $I = -10°$, $a_{95} = 4.6°$]. Thus the authors believe the magnetization to be primary. The inclination is between 10° and 14°, suggesting that the SCB was located near the equator during the Early-Middle Devonian (paleolatitude 7° north or south). Other Middle-Late Devonian data are from limestones of the Longmen Shan (Latitude = 31.9°N, Longitude = 104°E) in Sichuan Province, southwest SCB [*Wang*, 1986]. The Middle Devonian results have dual-polarity, yielding a mean inclination of -12° or a paleolatitude of 6° (north or south, $D = 309°$, $a_{95} = 9.9°$). The Late Devonian mean inclination for the same study is 18° or a paleolatitude of 9° (north or south, $D = 322°$, $a_{95} = 9.9°$). Although no fold test is available on the Devonian results, the results from overlying Carboniferous strata which have dual polarity pass the fold test. More recently, *Fang et al.* [1989] reported a Devonian pole from eastern Yunnan Province which gives a mean direction of $D = 279°$, $I = 3°$, $a_{95} = 6.9°$, giving a paleolatitude of 1.5°. Based on these results, we believe that the SCB was located in an equatorial region during the Devonian.

Discordant Devonian poles support substantial post-Devonian relative motions between Tarim, Siberia and Kazakhstan (Figure 13). Difference in shape between the Late Paleozoic APW paths of Tarim and Kazakhstan suggest that they were separate tectonic units during Devonian and Carboniferous times. The Late Paleozoic APW path of Tarim is similar to that of Siberia (Figure 13) despite a shift. The separation of APW paths from Tarim and Siberia suggest that Tarim has moved northeastward with respect to Siberia. It is very likely that the collision between India and Tibet is the main cause of this motion, though Tertiary paleomagnetic poles from both Tarim and Siberia are needed to test this hypothesis.

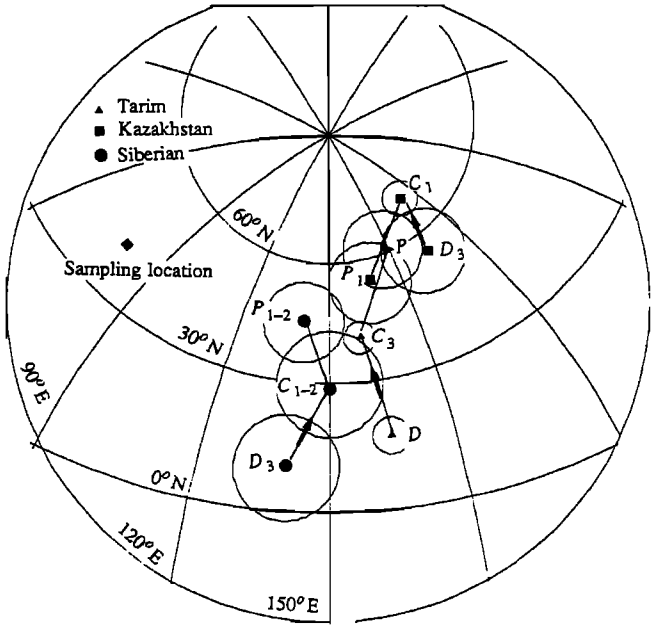


Fig. 13. Oblique polar projection of Late Paleozoic paleomagnetic poles for the Tarim, Siberian and Kazakhstan Blocks. Tarim's poles are on the basis of this study, *Zhai et al.* [1988] and *Sharps et al.* [1989]. The poles of the Kazakhstan and Siberian Blocks are from *Khramov et al.* [1981] and *Khramov et al.* [1985].

PALEOGEOGRAPHIC RECONSTRUCTION

In previous paleogeographic reconstructions, Tarim was either attached to the North China Block or was not discussed [*Ziegler*, 1981; *Scotese*, 1984; *Zonenshayn and Gorodniskiy*, 1977]. The difference in stratigraphic history and Permian/Carboniferous paleomagnetic data from the Tarim and North China Blocks indicate that these two blocks were separate during the Late Paleozoic.

A recent Late Devonian reconstruction of Gondwana and Asia made by *Long and Burrett* [1989] placed Tarim in a southerly latitudes of 6° ± 7°. Although no reference was cited for the paleomagnetic data used, we know only two reported Devonian paleomagnetic data from Tarim (*Bai et al.* [1987] and this study). Neither our results nor these of *Bai et al.* [1987] give a southern hemisphere position for Tarim during Late Devonian.

Our paleomagnetic data, combined with data published by others presented here and elsewhere [*Fang et al.*, 1989; *Wang*, 1986; *Tan and Zhai*, 1988; *Khramov et al.*, 1981; *Khramov et al.*, 1985] suggests that both the Tarim and South China Block were located in an equatorial region, far to the south of the Kazakhstan and Siberia Blocks. This view is supported by paleontologic evidence [*Gao and Wu*, 1984; *Smith*, 1988]. The Ordovician nautiloid *Endoceras* is found on both Tarim and South China Blocks. Traces of *Sinoceras chinese*, *Chisiloceras, Dicleroceras, and Vaginoceras* are present in the Keping and Kuruktag regions of the Tarim Block and in Guizhou, the Yangtze Gorges, Hunan and Sichuan of the South China Block. In addition, the Middle Ordovician *Yohophyllum* has been observed from Guizhou (South China) and Kuluktag (Tarim). During the Carboniferous and Early Permian, both the Tarim and South China Blocks belonged to the Cathaysia faunal province [*Smith*, 1988]. The barrier to exchange of terrestrial floras between the Tarim and Kazakhstan/Siberia Blocks started to break down in the Late Permian [*Smith*, 1988; *Hu*, 1985]. The Permian paleolatitude of Tarim is 33° N, whereas both the North China and South China Blocks are still located in equatorial regions during Late Permian times [*Zhao and Coe*, 1987]. Thus Tarim must have arrived to the southern margin of Asia earlier than the North and South China Blocks.

CONCLUSION

Paleomagnetic directions isolated from samples of 47 sites from the Subasi and Acha sections are consistent, with dual magnetic polarities at Subasi. These directions are believed to be primary Devonian magnetizations on the basis of the positive baked contact test and the presence of normal and reversed polarities within the same continuous stratigraphic section. Detailed thermal/AF demagnetization, rock magnetic study, observation of the overall APW path from Tarim and stratigraphic correlation lead us to believe that the results from the Yingan section are contaminated at least partially by Permian igneous activity. On the same grounds, we consider that the Devonian and Carboniferous results from the Sishichang section, 20 km east of the Yingan section, reported by *Bai et al.* [1987], are also overprinted by Permian igneous activity.

21562202b, 1990, B12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB12p19185 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Our data from Subasi reveal a 395 m reversed polarity zone overlying a 1603 m normal polarity zone. The maximum inclination change from the bottom to the top of the section is 24°, with younger beds yielding steeper inclination.

The Late Paleozoic APW path from Tarim suggests that it was attached to a subducting plate, and moved from an equatorial region during Devonian to 33°N during Permian. During the Devonian, Tarim was separated from the Angaran plate to the south, but was in the same paleolatitude as the South China Block, reconciling the fact that these two blocks shared the same floras during at least most of the Paleozoic.

APPENDIX

Natural remanent magnetization (NRM) of a specimen containing a three-component remanence can be written as

$$\mathbf{J}_0 = a_{01}\mathbf{v}_1 + a_{02}\mathbf{v}_2 + a_{03}\mathbf{v}_3$$

where $\mathbf{J}_0$ is the total NRM; $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$ are unit vectors in the direction of each of three components constituting the NRM direction; a_{01}, a_{02} and a_{03} are scaling factors for the three components, respectively. During an n step demagnetization, the total magnetization at each demagnetization step can be written as

$$\begin{aligned}
\mathbf{J}_1 &= a_{11}\mathbf{v}_1 + a_{12}\mathbf{v}_2 + a_{13}\mathbf{v}_3 \\
\mathbf{J}_2 &= a_{21}\mathbf{v}_1 + a_{22}\mathbf{v}_2 + a_{23}\mathbf{v}_3 \\
&\ \ \vdots \\
\mathbf{J}_{n-1} &= a_{n-11}\mathbf{v}_1 + a_{n-12}\mathbf{v}_2 + a_{n-13}\mathbf{v}_3 \\
\mathbf{J}_n &= a_{n1}\mathbf{v}_1 + a_{n2}\mathbf{v}_2 + a_{n3}\mathbf{v}_3
\end{aligned}$$

or in matrix form

$$\begin{bmatrix} \mathbf{J}_0 \\ \mathbf{J}_1 \\ \vdots \\ \mathbf{J}_{n-1} \\ \mathbf{J}_n \end{bmatrix} = \begin{bmatrix} a_{01} & a_{02} & a_{03} \\ a_{11} & a_{12} & a_{13} \\ \vdots & \vdots & \vdots \\ a_{n-11} & a_{n-12} & a_{n-13} \\ a_{n1} & a_{n2} & a_{n3} \end{bmatrix} \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \mathbf{v}_3 \end{bmatrix}$$

which can be also written in the following form:

$$\mathbf{J} = A\mathbf{V}$$

where $\mathbf{J}$ are total magnetizations measured at each demagnetization step; $\mathbf{V}$ are magnetization components derived from principal component analysis [*Kirschvink*, 1980]; A is a component intensity spectra matrix for which we wish to solve.

A is obtained by a simple matrix inversion:

$$\mathbf{J}\mathbf{V}^T = A\mathbf{V}\mathbf{V}^T$$

$$A = (\mathbf{J}\mathbf{V}^T)(\mathbf{V}\mathbf{V}^T)^{-1}$$

Where $\mathbf{V}^T$ is the transpose matrix of $\mathbf{V}$, $(\mathbf{V}\mathbf{V}^T)^{-1}$ is the inverse matrix of $(\mathbf{V}\mathbf{V}^T)$ which is a square matrix. The plot of matrix A as a function of demagnetization steps is a component intensity spectrum (a computer program written in the C language to perform component intensity analysis is available from Y. Li upon request).

Acknowledgments. We thank Shi Guangqian, Tan Chengzhe, Zheng Ming and Susan McWilliams for help with sample collection. Financial support was provided in part by the National Science Foundation (EAR-8618562, EAR-8718562 and EAR-881774), the Comprehensive Research Project Office for Accelerating the Investigation of Mineral Resources of Xinjiang, China, National Science Foundation of China and the Stanford-China Geosciences Industrial Affiliates Program.

REFERENCES

Bai, Y., G. Chen, Q. Sun, Y. Sun, Y. Li, Y. Dong, and D. Sun, Late Paleozoic polar wander path for the Tarim Block and tectonic significance, *Tectonophysics, 139*, 145-153, 1987.

Carroll, A., S. Graham, M. Hendrix, J. Chu, C. McKnight, C. Liang, and X. Xiao, Late Carboniferous collision of the Tarim craton, NW China (abstract), *Geol. Soc. Am. Abstr. with Programs, 21*, A348, 1989.

Chang, C. et al., Preliminary conclusions of the Royal Society and Academia Sinica 1985 geotraverse of Tibet, *Nature, 323*, 501-507, 1986.

Chen, Z., and Y. Liang, Research on several problems of structural geology in Tianshan, Xinjiang (in Chinese), *Xinjiang Geol., 3*, 1-13, 1985.

Cheng, S., G. Wang, S. Yang, J. Jin, and J. Zhu, The paleoplate tectonic of Xinjiang (in Chinese), *Xinjiang Geol., 4*, 1-26, 1986.

Fang, W., R. Van der Voo, and Q. Liang, Devonian paleomagnetism of Yunnan Province across the Shan Thai-south China suture, *Tectonics, 8*, 939-952, 1989.

Fisher, R. A., Dispersion on a sphere, *Proc. R. Soc. London Ser. A, 217*, 295-305, 1953.

Gao, Z., and S. Wu, Tectonical development of Precambrian-Paleozoic Tarim Massif, *Kexue Tongbao, 29*, 1723-1726, 1984.

Gordon, R., A. Cox, and S. O'Hare, Paleomagnetic Euler poles and the apparent polar wander and absolute motion of North America since the Carboniferous, *Tectonics, 3*, 499-537, 1984.

Harland, W. B., A. V. Cox, P. G. Llewelyn, A. G. Pickton, A. G. Smith, and R. Walter, *A Geologic Time Scale*, 131 pp., Cambridge University Press, New York, 1982.

Hu, Y., Characteristics of the Permian floras in the western part of China (in Chinese), *Reg. Geol. China, 12*, 100-108, 1985.

Ji, X., and P. J. Coney, Accreted terranes of China, in *Tectonostratigraphic Terranes of the Circum-Pacific Region*, pp. 349-362, Circum-Pacific Council for Energy and Mineral Resources, Houston, 1985.

Khramov, A. N., G. N. Petrova, and D. M. Pechersky, Paleomagnetism of the Soviet Union, in *Paleoreconstruction of the Continents, Geodyn. Ser., vol. 2*, edited by M. W. McElhinny and D. A. Valencio, pp. 159-194, AGU, Washington, D. C., 1981.

Khramov, A. N., E. L. Gurevich, R. A. Komissarova, E. P. Osipova, S. A. Pisarevsky, V. P. Rodionov, and I. P. Slautsitais, Paleomagnetism, microplates and Siberian Plate consolidation, *J. Geodyn., 2*, 127-139, 1985.

King, R. F., The remanent magnetism of artificially deposited sediments, *Mon. Not. R. Astron. Geophys. Suppl., 7*, 115-134, 1955.

Kirschvink, J. L., The least-squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc., 62*, 699-718, 1980.

Li, Y. A., Q. Li, Y. Liu, Z. Zhang, and X. Jing, Study on Paleomagnetism of Tarim Craton since Late Paleozoic (in Chinese), *Xinjiang Geol., 7*, 1-77, 1989.

Li, Y. P., Paleomagnetism of western China and the southern Sierra Nevada, Ph.D. dissertation, 234 pp., Stanford Univ., Stanford, Calif., 1988.

Li, Y. P., Y. A. Li, R. Sharps, M. McWilliams, Z. Gao, Sinian paleomagnetic results from the Tarim Block, western China, *Precambrian Res.*, in press, 1990.

Li, Y. P., M. McWilliams, A. Cox, R. Sharps, Y. A. Li, Z. Gao, Z. Zhang, and Y. Zhai, Late Permian paleomagnetic pole from dikes of the Tarim craton, China, *Geology, 16*, 275-278, 1988.

Long, J., and C. Burrett, Fish from the Upper Devonian of the Shan-Thai terrane indicate proximity to east Gondwana and south China terranes, *Geology, 17*, 811-813, 1989.

McFadden, P., X. Ma, M. McElhinny, and Z. Zhang, Permo-Triassic magnetostratigraphy in China: Northern Tarim, *Earth Planet. Sci. Lett., 87*, 152-160, 1988.

Molnar, P., and P. Tapponnier, Cenozoic tectonics of Asia: Effects of a continental collision, *Science, 189*, 419-425, 1975.

Nagata, T., *Rock Magnetism*, 2nd ed., 351 pp., Plenum, New York, 1961.

Nie, S., D. B. Rowley, and A. M. Ziegler, Constraints on the locations of Asian microcontinents in Palaeo-Tethys during the Late Paleozoic, in *Palaeozoic palaeogeography and biogeography*, edited by W. S. McKerrow and C. R. Scotese, Geol. Soc. London Memoir, in press, 1990.

Ren, J., C. Jiang, Z. Zhang, and D. Qin, *Geotectonic Evolution of China*, 203 pp., Springer-Verlag, New York, 1987.

Scotese, C. R., An Introduction to this volume: Paleozoic paleomagnetism and assembly of Pangea, in *Plate Reconstruction From Paleozoic Paleomagnetism, Geodyn. Ser. vol. 12*, edited by

21562202b, 1990, B12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB12p19185 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

R. Van der Voo, C. R. Scotese, and N. Bonhommet, pp.1-10, AGU, Washington, D. C., 1984.

Sharps, R., M. McWilliams, Y. P. Li, A. Cox, Z. Zhang, Y. Zhai, Z. Gao, Y. A. Li, and Q. Li, Lower Permian paleomagnetism of the Tarim Block, northwestern China, *Earth Planet. Sci. Lett., 92*, 275-291, 1989.

Smith, A. B., Late Paleozoic biogeography of East Asia and palaeontological constraints on plate tectonic reconstructions, *Philos. Trans. R. Soc. London Ser. A, 326*, 189-227, 1988.

Tan, C., and Y. Zhai, A study of early-middle Devonian paleomagnetism for southern Jiangsu, China (abstract), *Tectonics East. Asia West. Pac. Contin. Margin (DELP), 0155*, 59, 1988.

Wang, H., and B. Liu, Paleogeographic Frame of China since Middle Proterozoic (in Chinese), *J. Stratigraphy, 5*, 77-89, 1981.

Wang, Z., Method of analysis of remanent magnetization and study of Paleozoic paleomagnetic characters and their geologic implication of several geologic units in China (in Chinese), M.S. thesis, 119 pp.; China Geosci. Univ. at Beijing, 1986.

Watson, M. P., A. B. Hayward, D. N. Parkinson, and Z. M. Zhang, Plate tectonic history, basin development and petroleum source rock deposition onshore China, *Mar. Pet. Geol., 4*, 205-225, 1987.

Xinjiang Stratigraphic Compiling Group, *Regional Stratigraphic Tables of Northwest China, Volume on Xinjiang*, 496 pp., Geological Publishing House, Beijing, 1981.

Zhai, Y., Z. Zhang, Y. A. Li, Q. Li, Y. P. Li, M. McWilliams, A. Cox, and R. Sharps, A study of Upper Carboniferous paleomagnetism for the Tarim Block (in Chinese), *Geoscience, 2*, 43-56, 1988.

Zhang, L., and N. Wu, The geotectonics and the evolution of Tianshan (in Chinese), *Xinjiang Geol., 2*, 1-14, 1985.

Zhang, Z., S. Wu, Z. Gao, S. Xiao, and H. Ba, Research of sedimentary model from Late Carboniferous to Early Permian Epoch in Kalpin region, Xinjiang (in Chinese), *Xinjiang Geol., 1*, 9-20, 1983.

Zhang, Z. M., J. G. Liou, and R. G. Coleman, An outline of the plate tectonics of China, *Geol. Soc. Am. Bull., 95*, 295-312, 1984.

Zhao, X., and R. S. Coe, Paleomagnetic constraints on the collision and rotation of North and South China, *Nature, 327*, 141-144, 1987.

Zhao, X., R. Coe, Y. Zhou, H. Wu, and J. Wang, New paleomagnetic results from northern China: Collision and suturing with Siberia and Kazakhstan, *Tectonophysics*, in press, 1990.

Ziegler, A. M., Paleozoic Paleogeography, in *Paleoreconstruction of the continents, Geodyn. Ser., vol. 2*, edited by M. W. McElhinny and D. A. Valencio, pp. 31-37, AGU, Washington, D. C., 1981.

Zijderveld, J. D. A., AC demagnetization of rocks, in *Method in Paleomagnetism*, edited by D. W. Collinson, K. M. Creer and S. K. Runcorn, pp. 254-286, Elsevier, New York, 1967.

Zonenshayn, L. P., and A. M. Gorodniskiy, Paleozoic and Mesozoic reconsctructions of the continents and oceans, 1, Early and Middle Paleozoic reconstructions, *Geotectonics, 11*, 83-94, 1977.

Z. Gao, Q. Li, Y. A. Li, Geology Research Institute, Geology Bureau of Xinjiang, Urumqi, China.

Y. P. Li, M. McWilliams, R. Sharps, Department of Geophysics, Stanford University, Stanford, CA 94305.

Y. Zhai, Beijing Graduate School, China University of Geosciences, Beijing, China.

Z. Zhang, Geology Institute, Chinese Academy of Geological Sciences, Beijing, China.

(Received January 15, 1990;
revised April 25, 1990;
accepted June 6, 1990.)

21562202b, 1990, B12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB12p19185 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License